Manuscript 14: Lean 4 and Arb certification record

Magnetic properties of the Eulerian profile at the special point $i/\sqrt{2}$, version 14.91

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1. Scope.

This record states, for each numbered result of manuscript 14, version 14.91, what is kernel-checked in Lean 4, which scalar values are enclosed by Arb at 2000 bits, which steps are classical inputs and which formalisation tasks remain open. Equation, section and result numbers are those of version 14.91. All certificates are in the main branch of the repository `https://github.com/ralfanop/Certifications_Eulerian_polar-curve_computations` at commit `38957482`, and were audited from 5 to 7 November 2026; each pull request (PR) below links its workflow runs. The Lean certificates use Lean `v4.35.0-rc3` with the pinned Mathlib dependency recorded in their workflows, compiled from source; every axiom audit lists only `propext`, `Classical.choice` and `Quot.sound`, with no `sorryAx` and no `Lean.ofReduceBool`. The numerical certificates use `python-flint 0.9.0`, exact rational arithmetic and Arb balls at 2000 bits; they certify the stated scalar quantities and do not prove infinite-dimensional operator identities.

2. Overview.

In Table 1, "yes" means that the stated content is certified, apart from the classical inputs listed in Section 4; "partial" means that part of it is, the rest being listed in Section 5.

Table 1. Certification status of the numbered results.

Result	Lean 4	Arb	Pull requests
Proposition 2.1 (special-value chain)	—	yes	PR 38
Corollary 2.2 (rational bounds)	—	yes	PR 38
Lemma 2.3 (uniqueness of z_0)	yes	yes	PR 44
Proposition 3.1 (Sonine)	partial	yes	PR 31, PR 38
Proposition 3.3 (modified Helmholtz)	partial	yes	PR 38, branch `cor36`
Proposition 4.1 (Langevin)	partial	yes	PR 38, PR 46
Theorem 4.2 (noninteracting magnet)	yes	yes	PR 40, PR 48, PR 61
Displays (54)–(55) (Riccati, Painlevé III)	—	yes	PR 61
Display (60) (n-vector)	$\lambda = 0$	$\lambda = 0, \pm\frac{1}{2}$	PR 40, PR 47
Definition 5.1 (Gram operator)	yes	—	PR 33
Theorem 5.2 (Gram commutation)	yes	—	PR 37
Corollary 5.3 (Eulerian specialisation)	partial	yes	PR 33, PR 38
Proposition 5.4 (spheroidal basis)	yes	—	PR 41, PR 42, PR 43
Proposition 6.2 (two determinants)	partial	yes	PR 44, branch `dirichlet-finite`
Proposition 6.3 (resolvent traces)	yes	yes	PR 44
Corollary 6.4 (Hilbert–Schmidt trace)	yes	yes	PR 44
Proposition 6.5 (zeta expansion)	—	yes	PR 39
Theorem 7.1 (modular duplication)	partial	yes	PR 44
Theorem 8.1 (quadratic covering)	partial	yes	PR 45
Proposition 8.2 (Montgomery–Taylor)	yes	yes	PR 45, PR 61

Result	Lean 4	Arb	Pull requests
Proposition 8.4 (self-dual coupling)	partial	yes	PR 45
Proposition 8.6 (paths and Viète)	partial	yes	PR 45, PR 61
Proposition 8.7 (quartic minimal polynomial)	partial	yes	PR 61
Proposition 8.8 (lattice walks)	partial	yes	PR 45
Lemma 9.2 (time reversal)	yes	yes	PR 45
Theorem 9.3 (Bloch antipode)	partial	yes	PR 45, PR 46
Corollary 9.4 (averages over pure states)	partial	yes	PR 45, PR 46
Proposition 9.5 (Brillouin splitting)	yes	yes	PR 44, PR 45
Proposition 9.7 (involutions)	partial	yes	PR 45
Theorem 10.1 (Eulerian–Parry)	yes	yes	PR 30

3. Content of the certificates.

Sections 2–3. The script `special_point_i_sqrt2_arb.py` encloses $r_E(z_0) = \sinh(a)/a$ and checks, in exact rational arithmetic, the bounds (19). The duplication identity (15) is kernel-checked in the `Bridge` library. The Sonine certificates (PR 31; CI 41853697588) prove the integral moments c_n for every index, their factorial formula, the interchange of the integral with absolutely convergent even series, and the spherical integral identities of the series-defined J_0 and I_0 at every complex argument; the radial-series certificate (CI 41852735785) proves the intertwining for coefficient-defined radial operators under its summability hypothesis. The differential equation (36) is kernel-checked in six declarations on the branch `manuscript14-cor36-lean-arb-20261104` (CI 41824974185).

Section 4. The values (52)–(53) are enclosed and proved correctly rounded (CI 42207399251). The module `BlochMeasureCertificate` (PR 46) proves $\int e^{x\omega_3}\,\mathrm{d}\mu = \sinh x/x$ with ω_3 uniform on $(-1,1)$, which gives the normalisation (38) through the coordinates of the first input of Section 2.2. The module `Theorem42Certificate` (PR 48; CI 42356565741; twenty-one audited statements) proves Theorem 4.2: for (i), $u\,\partial_u \log B(u) = [1 + L(H/2)]/2$, its derivative $L'(H/2)/4$ in H, the mean of the quasi-power $B(u)^{n+1}$, and the mean and variance of any finite law tilted by u^k as the first two derivatives of its logarithmic normalisation; for (ii), the probabilities $1/2$, $1/6$ and $1/4$ of a descent, of two adjacent descents and of two descents at distance at least two, for every n, hence (46), the mean $(n-1)/2$, (47) and (50); for (iii), the factorisation into independent spins with the fields (51); and (iv). The script of PR 40 (CI 42239016609; 145 checks) certifies (46)–(47) and all descent correlations for $n \le 9$, (47) for $n \le 670$, (50) with its coefficients, (57)–(59) at a and b, the real zeros and (51) for $n \le 18$, and (60) at $\lambda = \pm 1/2$. The module `PlanarSpinCertificate` (PR 47; CI 42355540276; eight audited statements) proves (60) at $\lambda = 0$: the arcsine law ν_0 is the image of the uniform angle under cos, its moments are $\binom{2m}{m}/4^m$ and 0, and its transform is $\sum_k (x^2/4)^k/(k!)^2$, also as an average over the full circle; its script (CI 42355345132; 13 checks) compares the series with I_0 at ten points, among them $I_0(z_0) = J_0(1/\sqrt{2})$, and with the integrals.

Section 5. CI 41840633678 checks seven declarations on the complex-L^2 Gram integral, its exact parameter and measure, the nuclear decomposition and the basis-independent trace, and CI 41841459971 thirteen kernel and graph-domain lemmas; the kernel balance, the core-extension criterion, the Gegenbauer spectrum and the trace formula (64) are in the `GegenbauerFock` library. The module `GramCommutationTheorem52` (PR 37; CI 42209958304) proves Theorem 5.2 for every $c > 0$, the inclusion $\mathcal{S}_{1/2,c}\mathcal{H}_{1/2} \subset \mathrm{Dom}(\mathcal{A}_c)$ and (63), for the graph closure of the polynomial Legendre operator perturbed by M_{ξ^2}. The module `GegenbauerSpectralRealisation` (PR 42;

CI 42260542889) identifies this closure with (8)–(9), proves it selfadjoint with compact resolvent, and proves that $\mathcal{A}_c + c^2 + 1$ has a compact inverse. The module `SpheroidalProposition54` (PR 41; CI 42249877445) proves Proposition 5.4: parity, simplicity within each parity class (from the moments rather than from (74)), positivity of the $\mathcal{S}_{1/2,c}$-eigenvalues and a joint Hilbert basis with $\sum_n \sigma_n = {}_1F_1(1/2; 3/2; c)$. The module `SpheroidalDLMFIdentification` (PR 43; CI 42272154201) identifies the basis vectors with the spheroidal functions $\mathrm{Ps}_n^0(\cdot, -c^2)$, the definition of DLMF §§30.2–30.3 entering as an explicit hypothesis.

Section 6. The branch `manuscript14-dirichlet-finite-20261104` (CI 41824926567, CI 41825497267) checks fourteen declarations on the finite truncations of the products and traces, with thirteen Arb comparisons. The module `Section67ScalarCertificate` (PR 44; CI 42283397911; nineteen audited statements) proves the Euler products (13)–(14) as the products (79)–(80) over the spectra (76) and (78), with (81), the spectral sums (82)–(84) and (89)–(90) for every $x > 0$, the values (85) and (87), and (91)–(92). The script of PR 39 (CI 42224556937; 92 checks) certifies Proposition 6.5: the Bernoulli numbers and the rational terms of (97), the values $\zeta(2m)$ with exact tail bounds, the termwise agreement of the two forms of (94), and the enclosure of $\log r_E(z_0)$.

Section 7 and Lemma 2.3. The same module proves Lemma 2.3 (uniqueness of z_0 in the upper half-plane, $S(z_0) = z_1 = 2z_0$, $S(z_1) = z_0$ and (22)–(23)) and Theorem 7.1: (98), the product (99) over the spectrum of A_2 and its splitting into the products over the spectra of A and A_{DN}, (100) in product form, (101)–(103) and the companion values. Its script (CI 42282862670; 45 checks) certifies (21)–(103) as ball identities, the products and traces with rigorous tails, and the rounding of the values of Remark 7.4.

Sections 8–9. The module `Section89Certificate` (PR 45; CI 42299447127; thirty-three audited statements) proves: in Theorem 8.1, the fibres $\{\tau, -\tau\}$ of $\mathcal{Q}$, the deck transformations, (115) with r_E the limit of the partial products of (114), $m(-H) = 1 - m(H)$, the lifts of (116), the doubling of winding numbers, the fibres over $\pm 1/4$, (118) and the uniqueness of z_0; Proposition 8.2 ((121) from the Mittag-Leffler expansion of the cotangent, the logarithmic derivative, (123)); in Proposition 8.4, (126), (131)–(135), $4G_*^2 + 4G_* - 1 = 0 \Leftrightarrow \sinh 2K_{\mathrm{sd}} = 1$ and the arcsine moments; part (i) of Proposition 8.6, (136) with its zeros, the Perron eigenvalue, the eigenvector relation and (137); the two identities in the proof of Proposition 8.8; Lemma 9.2; $\omega_{\mathrm{Bl}} \circ \Theta = -\omega_{\mathrm{Bl}}$ and $|\omega_{\mathrm{Bl}}| = 1$; (160), (161) and the time reversal of the thermal state, the spectral forms (158)–(159) being (80) and (83) of PR 44; and parts (i), (iii) and (iv) of Proposition 9.7. The module `BlochMeasureCertificate` (PR 46; CI 42307987424; fifteen audited statements) proves that the image of μ under $\mathfrak{z} \mapsto (\omega_3, \arg \mathfrak{z})$, the axial coordinate and the azimuth of $\omega_{\mathrm{Bl}}(\mathfrak{z})$, is $(4\pi)^{-1} \mathrm{d}\omega_3 \, \mathrm{d}\varphi$ on $(-1, 1) \times (-\pi, \pi)$, that μ is a probability measure, (155) for every real $x \neq 0$ and (156), the mean $L(x)$ of ω_3 under $e^{x\omega_3} \mathrm{d}\mu$, the invariance of the law of ω_3 under $\mathfrak{z} \mapsto -1/\bar{\mathfrak{z}}$, and the bijection of $\mathbb{C}$ onto the sphere without its south pole. The script of PR 45 (CI 42299795864; 115 checks) certifies the rounding of (120), (125) and (127), the closed forms as ball identities, the traces of (121) and (123) with two-sided tail bounds, (122) in exact arithmetic, (128)–(130), the characteristic polynomials of the paths for $m \leq 12$, (138), (148) for $2n \leq 24$, (149)–(150) at $q = 1/8$, (155)–(156) as integrals, and (157) against the sublevel sums for $J = 1/2, 1, 3/2, 7/2$.

Additions of 7 November 2026 (evening). The script `numerics/arb/manuscript14_evening_additions_ar` (PR 61; CI 42422614198; 45 checks, exact rational and $\mathbb{Q}(\sqrt{2})$ arithmetic and Arb at 2000 bits) certifies: the Riccati equation (54) of Section 4.3 at six points, its general solution L_ϱ for four values of ϱ, the saddle (eigenvalues 1, -2, separatrix slope $1/3 = L'(0)$) and the node (eigenvalues 1, 2) of the Riccati foliation, $\mathsf{w}(z) = 1/z - \cot z$ with $\mathsf{w}(ix) = iL(x)$ and the third Painlevé equation (55) at four complex points; in the proof of Theorem 4.2, Knuth's Bernoulli numbers

($B_1 = 1/2$), the equality of the kernel (49) with the form of Borodin, Diaconis and Fulman, the generating function (48), the mean $(n-1)/2$ and the variance $(n+1)/12$ for $n \le 30$ and, with $\mathrm{k}(0) = -1/2$, the mean $-(n-1)/2$, the negative variance and the polynomials $3-x$ and $x^2-8x+13$, the Eulerian polynomials as $n!\det[I+(x-1)\mathcal{K}_{n-1}]$ for $n \le 7$, and Bernoulli's sums of powers with $B_1 = \pm 1/2$; the eight terms of (122) and their zeta form; the middle term of (138); and Proposition 8.7: the roots of (139), its irreducibility (Eisenstein at 2 for $\mu_{\mathrm{hex}}^4 - 4\mu_{\mathrm{hex}}^2 + 2$ and the criterion for biquadratics) and primitivity, so that it is the minimal polynomial of ϖ over $\mathbb{Z}$, (140), the integral (141), Bender's polynomial (142)–(145) for seven rational values of $\gamma_{\mathrm{PT}}^2 \in [0,2)$, (146) with $\cot(\pi/8) = 1+\sqrt{2}$, (147), and the midradius of the square antiprism with unit edges. The module `QuarticResidueCertificate` (PR 61; CI 42422614173; fourteen audited statements) proves the algebraic identities of (139), (140), (142)–(147) for $\varpi = \sqrt{2+\sqrt{2}}/\sqrt{8}$ and $\varpi' = 1/(2\sqrt{2+\sqrt{2}})$; the integral (141), the irreducibility and the value of $\cot(\pi/8)$ are certified only in the script.

Section 10. Theorem 10.1 is kernel-checked in full (CI 41812736721): the adjacent thresholds, the greedy digits, the rational orbit-defect bounds, the endpoint jump, the rational series of $\beta_E = \sqrt{2}\sinh(1/\sqrt{2})$ (module `EulerianSinhSeries`), its transcendence and the absence of eventual periodicity; the finite word at β_{31} is checked separately. The Hermite–Lindemann instance is compiled from source adapted, under its Apache 2.0 licence and with attribution, from Yuyang Zhao's Mathlib pull request #28013, not yet merged on 7 November 2026.

4. Classical inputs.

The following enter the certificates as explicit hypotheses or are not reproved: the inputs of Section 2.2 of the manuscript, among them the surface measure $\mathrm{d}\xi\,\mathrm{d}\varphi/(4\pi)$ of $\mathbb{S}^2$ in axial coordinates; the identification of the Fredholm determinant and trace of a positive trace-class operator with the product and sum over its spectrum; the quasi-power approximation $\mathrm{A}_n(u) = B(u)^{n+1}[1+O(2^{-n})]$ (Flajolet–Sedgewick, Example IX.37); the real, simple, negative zeros of A_n (Frobenius); the determinantal structure of the descent process (Borodin–Diaconis–Fulman, Theorem 5.1); the j-values of Lemma 2.3; the covering-space statements of Theorem 8.1(i) and the exact sequence (117); and the definition of the spheroidal functions in DLMF §§30.2–30.3.

5. Open formalisation tasks.

Not formalised in Lean: the identification of the radial operators with H_2, H_3 at the origin (Proposition 3.1); the uniqueness of the regular radial solution (Proposition 3.3); Proposition 4.1 on the sphere itself; the zeta-regularised determinants (Proposition 6.2); Proposition 6.5; the unitary splitting $UA_2U^{-1} = A \oplus A_{\mathrm{DN}}$ (Theorem 7.1); the boundary value (128) (Proposition 8.4); part (ii) of Proposition 8.6 for general m, and (138); the operator statements of Proposition 8.8; the extension of ω_{Bl} to $\mathfrak{z} = \infty$ and the homeomorphism of Theorem 9.3; the invariance of μ itself under Θ (Corollary 9.4). For Proposition 2.1 and Corollary 2.2 this record lists only the numerical certificates. The data of the figures of Section 4, computed from the exact Eulerian numbers up to $n = 670$, carry no certificate.

6. Numerical jobs.

The workflow `manuscript14-arb-2000bit.yml`, introduced by PR 38, runs the fourteen numerical scripts of the manuscript at 2000 bits: 572 checks in all, of which 162 in the nine original scripts (63 of the register of 4 November 2026, 35 Bessel, 15 Gram and resolvent, 13 Dirichlet, 28 Sonine and modified Helmholtz, 8 for Proposition 4.1), 92 for Proposition 6.5, 145 for Section 4.6, 45 for Sections 6–7, 115 for Sections 8–9 and 13 for (60) at $\lambda = 0$. The exact rational truncation bounds are those of the original scripts; only the ball precision was raised.